Booklet Questions:-

Ex6:- [B.Q:-1]:- B & A are 180 km apart. They started simultaneously towards each other, with speeds of 10kmph & 20kmph respectively. In how many hr's will they meet?

Ans:- B ——— A
→ 10kmph, 180km, 20kmph

$$t_{meet}\,[\text{Opposite direction}] = \frac{\text{dist b/w them}}{\text{Relative speed (+)}}$$

$$= \frac{180}{10+20} = 6 \text{ hr's}$$

After 6 hr's of start, they meet each other.

Ex:7:- [B.Q:-3]:- A man travelled a distance of 61 km in 9 hr's. He travelled partly on foot @ 4kmph & partly on bicycle @ 9kmph. What is the distance travelled on foot?

Ans:- Let's consider dist travelled on foot = x

then " " " bicycle = $61-x$ (Total - foot)

Total time = $\text{time}_{foot} + \text{time}_{bicycle}$.

$$9 = \frac{x}{4} + \frac{(61-x)}{9} \Rightarrow \frac{9x+4(61-x)}{36} = 9$$

$$\Rightarrow \frac{9x+244-4x}{36} = 9 \Rightarrow 5x+244 = 36\times 9$$

$$\Rightarrow 5x = 324-244 = 80 \Rightarrow \boxed{x = 16\text{km}}$$

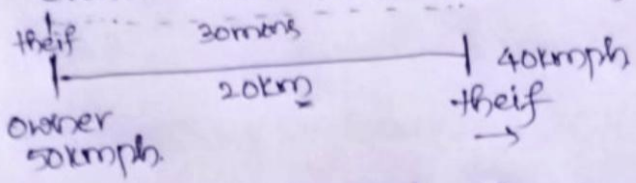
Ex:8:- [B.Q:-5]:- A man goes to his office from his house @ 3kmph & returns @ 2kmph. If he takes total 5 hr's. The dist b/w office & house?

Ans:- Total time = $T_{going} + T_{return} \Rightarrow 5 = \frac{d}{3} + \frac{d}{2}$

$$\Rightarrow 5 = \frac{2d+3d}{6} \Rightarrow \boxed{d = 6\text{ km}}$$

Ex:-9:- [B.Q:-7]:- A theif goes away with a car @ 40kmph. The theft has been discovered after half an hour & the owner sets off on a bike @ 50kmph. When will the owner over take the theif?

Ans:-



Theif started & observed after 30mins, meanwhile he travelled 20km in 30mins with 40kmph speed. So, when owner observed the dist b/w owner & theif is 20km.

$$t_{overtake} = \frac{\text{dist b/w them}}{\text{Relative Speed (same direction)}}$$

$$= \frac{20}{50-40} = \frac{20}{10} = 2 \text{ hr's}$$

After 2 hr's of owner started, overtakes the theif.

Ex:-10:- [B.Q:-8]:- If a person walks @ 14kmph instead of 10kmph, he would have walked 20km more. what is the actual distance travelled by him?

Ans:- 14kmph instead of 10kmph means 4km for every hour he is covering 20km more means.

4km → 1 hr
20km → $\frac{20 \times 1}{4}$ = 5 hr's of journey.

Actual dist travelled = 5 hr's x original speed (10kmph)
= 50 km

Ex:-11:- [BQ:-9]:- The ratio of speed of 2 cars is 7:8. 2nd car covers 400km in 4hr's. The speed of 1st car is?

Ans:- 8 parts = $\frac{400}{4}$ = 100 kmph ⇒ 1 part = $\frac{100}{8}$ = 12.5 kmph

1st car speed = 7 parts = 7 x 12.5 = 87.5 kmph